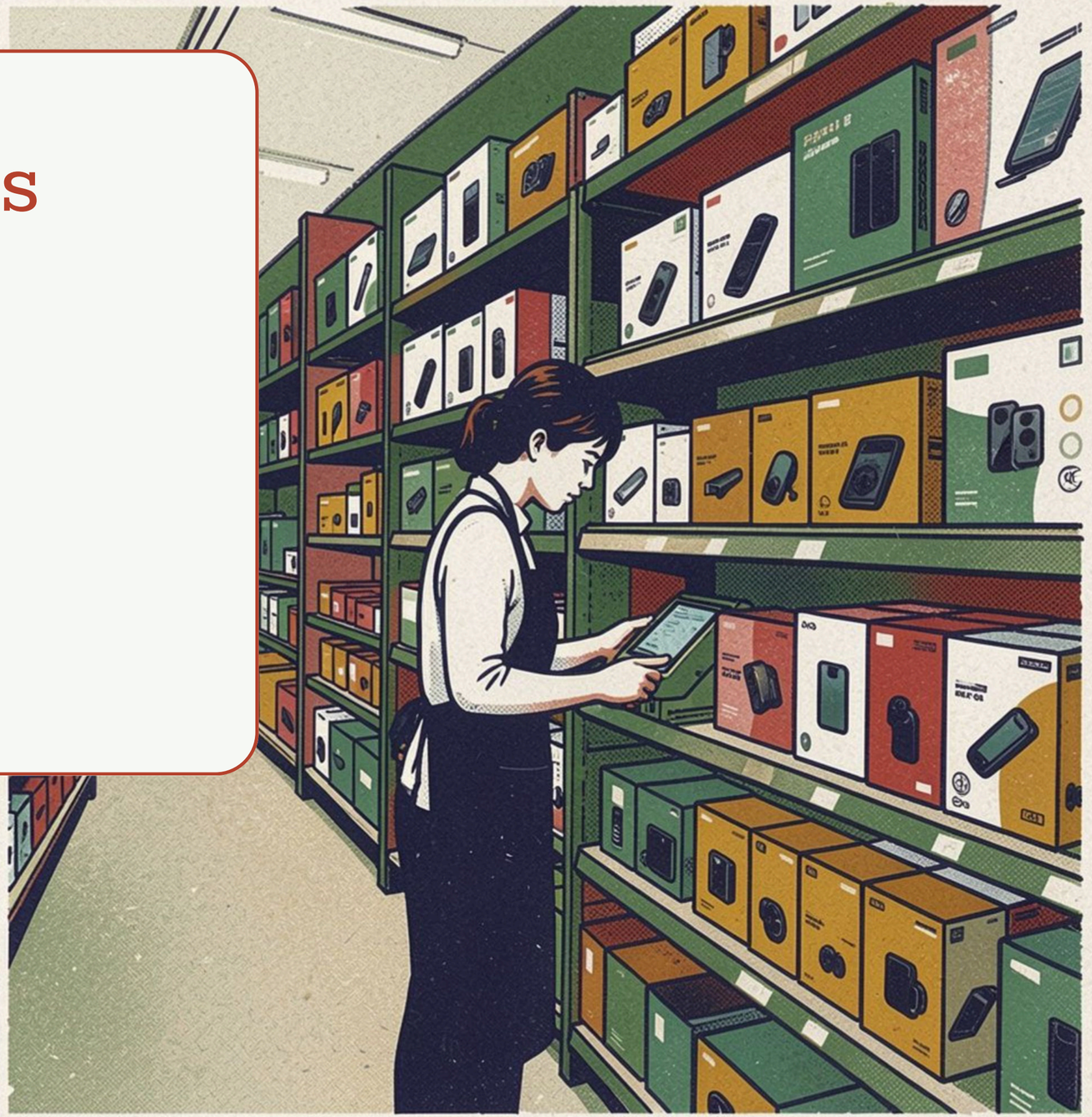
Retail Sales Analytics

From raw transactions to practical performance decisions.

Tools: Python | MySQL | Power BI
Dataset: Superstore Sales Dataset



DATASET OVERVIEW

A four-year dataset turns retail activity into a decision system

Superstore Sales Dataset

9,994

records

2011–2014

four years

3

product categories

4

regions

Objective: analyze sales, profit, products, regions, and customer segments.

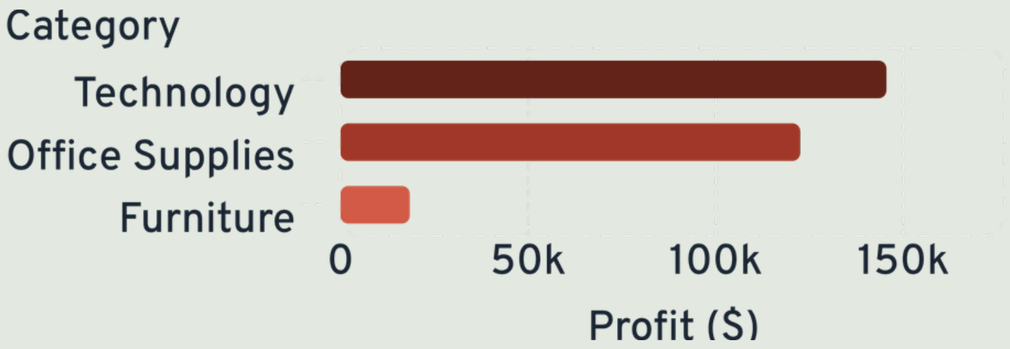


PYTHON · PANDAS

Python moves from raw rows to trend-ready analysis

Use Python and Pandas for loading, exploration, and analysis.

1. Load data
2. Explore data
3. Analyze trends
4. Create visualizations



Profit by category supplied values

Monthly sales trend

Labelled placeholder: exact monthly values were not supplied.

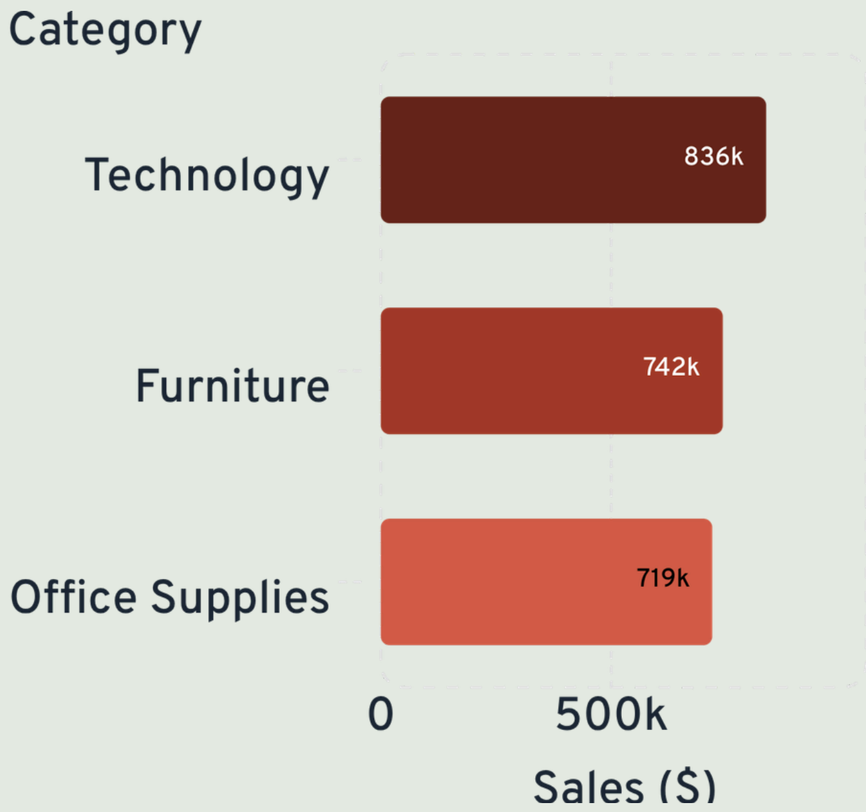
MYSQL · CATEGORY SALES

Technology leads category sales at $836K

MySQL answers focused business questions with repeatable queries.

SAMPLE QUERY

```
SELECT Category, SUM(Sales) AS total_sales
FROM superstore GROUP BY Category
ORDER BY total_sales DESC;
```



$2.30M in sales generated $286K in profit

$2.30M

Total Sales
$2,297,200.86

$286K

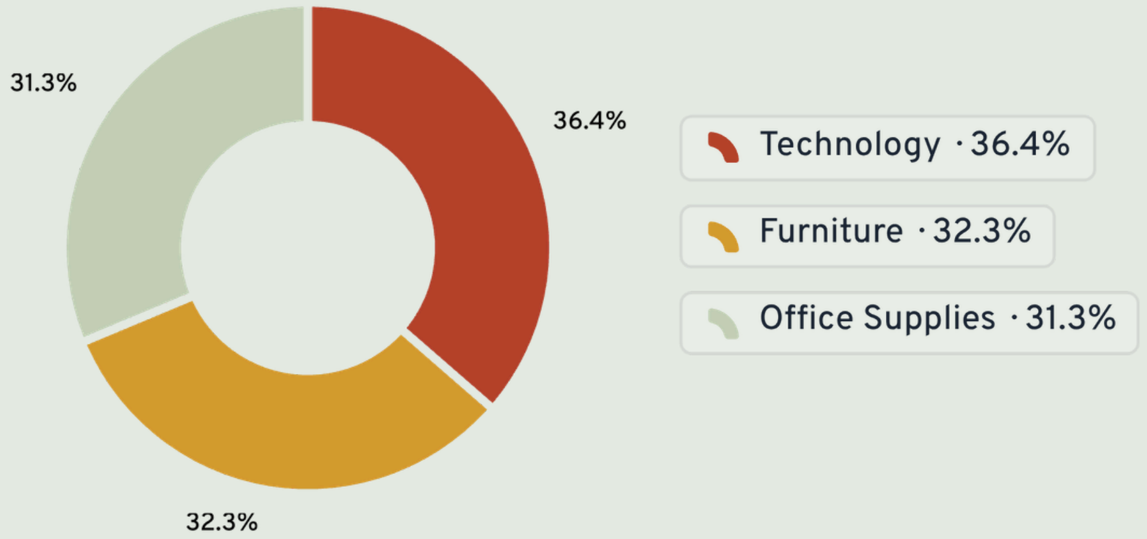
Total Profit
$286,397.02

5,009

Total Orders

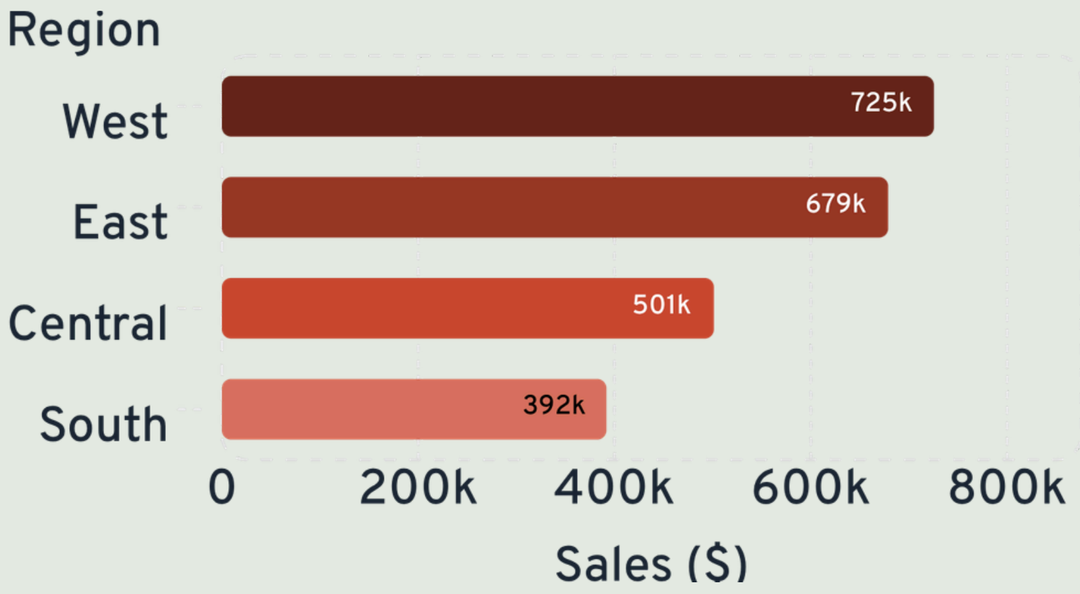
37,873

Total Quantity

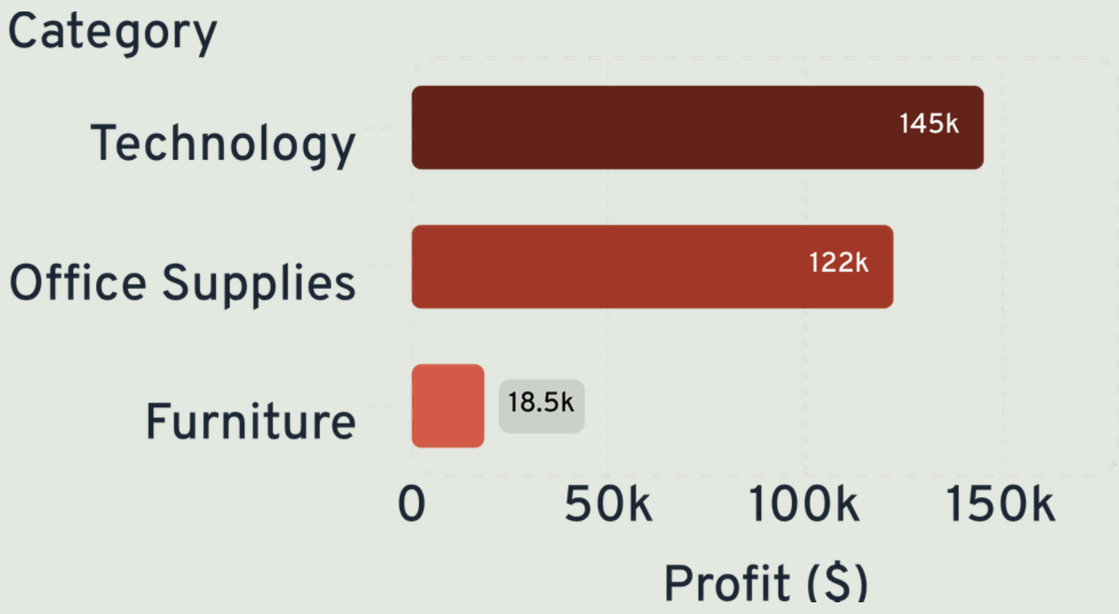


PROFITABILITY · REGIONAL SALES

Furniture sells meaningfully but returns far less profit



West leads regional sales at $725,457.82.



Furniture profit is much lower despite substantial sales.

POWER BI ·PERFORMANCE VIEW

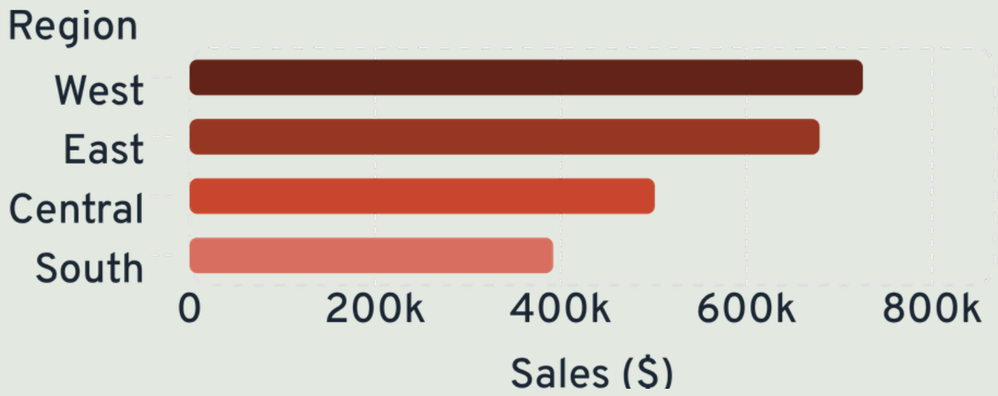
The Power BI dashboard brings the analysis together



KPI Cards
Monthly Sales Trend
Sales by Category
Profit by Region
Sales by Customer Segment
Year, Region, and Category Slicers

KEY FINDINGS

Technology and the West region set the pace



West leads the four regions; overall sales were approximately $2.30 million.

91.9%

Below the assumed $2,500,000 target.

Technology

Highest total sales and profit.

Furniture

Lowest profit among the three categories.

RECOMMENDATIONS

Four actions can turn analysis into performance

01

Review Furniture pricing, costs, and discounting to improve profitability.

02

Study the West region's performance to identify practices worth replicating.

03

Monitor monthly sales trends to support business planning.

04

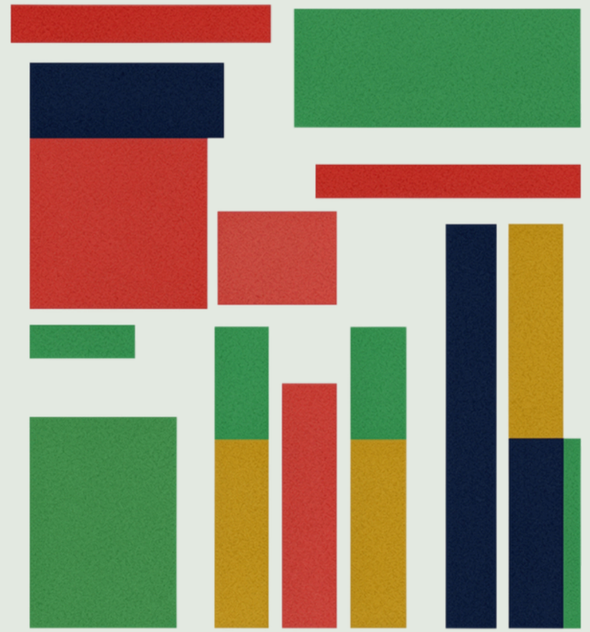
Use Power BI to track KPIs and compare categories and regions.

CONNECTED WORKFLOW

Data becomes decisions through one connected workflow

Data → Analysis → Insights → Dashboard

1. Python explores the dataset.
2. MySQL answers business questions.
3. Power BI makes performance visible.



Thank You